

PV–PORT 135 A port board

Rev F0 | 2026–10–04 | Schematics authored by Chinmoy Bhuyan | Ports A (PV) and B (bus/battery) of PV–P75, common negative BUS–

01 Connectors and sensor supply

PORT 26–way to CTRL–C2000, 14–way to SYS–IO–AUX, ID + hold readback, +24V → +3V3S buck, AGND tie, PE bond

File: sheets/01_io.kicad_sch

02 Port A power path

Terminals, fuses, SPD, EMI filter, contactor, precharge relay, coil drives, hold stage, shunt, discharge + time limit, bleeder

File: sheets/02_portA_power.kicad_sch

03 Port A sensing and hold logic

VA bus side, VAX terminal side (AMC3330), IA current (AMC3302), two–channel hold logic, proof–test injection, readback RB1/RB2

File: sheets/03_portA_sense.kicad_sch

04 Port B power path

Terminals, fuses, SPD, EMI filter, contactor, precharge relay, coil drives, hold stage, shunt, discharge + time limit, bleeder

File: sheets/04_portB_power.kicad_sch

05 Port B sensing and hold logic

VB bus side, VBX terminal side (AMC3330), IB current (AMC3302), two–channel hold logic, proof–test injection, readback RB1/RB2

File: sheets/05_portB_sense.kicad_sch

06 Insulation measurement

AUX1 = V(PE–BUS–), AUX2 = V(PE–A+) (AMC3330); 992 k test strings A+ → PE, BUS– → PE, each with a common–source SiC pair (TPSI3050)

File: sheets/06_imd.kicad_sch

Common negative: port A and port B share BUS– (non–isolated buck–boost), so both ports are one HV domain; the main contactors switch the + pole only, fuses are in both poles. Chassis–mounted parts (terminals, fuses, CM choke, contactors, RST 200 resistors, thermostats, shunt) are drawn here with their real MPN or an RFQ / CUSTOM specification.

Isolators (checked on every build): AMC3330/AMC3302, TPSI3050–Q1, CNY65B, contactor and relay coil–contact. Declared HV–PE crossings: MOV3 of each varistor Y, Y capacitors, the PE–side SiC of each insulation–measurement pair, the PE–end element of the AUX dividers.

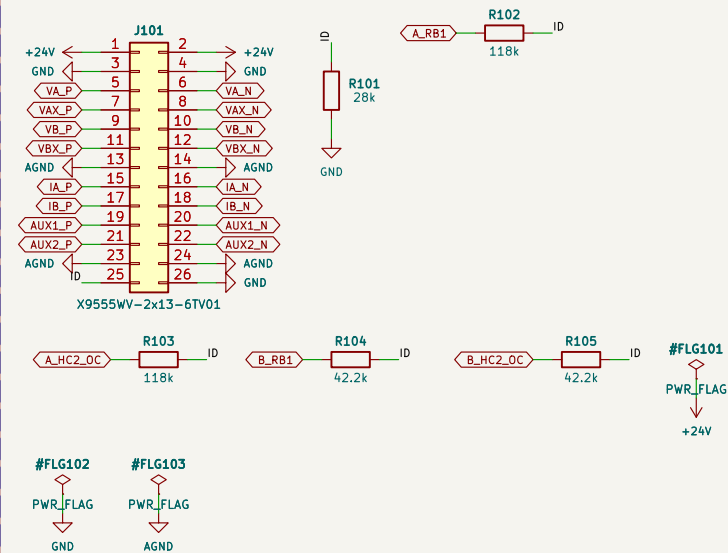
IC–17/IC–18: the whole board is conformal–coated to PD1 (IEC 60664–3 Type 1) over all HV copper and every HV–PELV isolator footprint (8–8.5 mm packages < 10 mm reinforced at PD2). Only the NH fuse links / bases and the DC terminals sit in the cooling–air path (PD3); the board, varistor network and branch fuses are in the closed PD2 compartment.

Surge credit (D–032, IC–01/02/03): the on–board varistor Y limits pole–PE and pole–pole to 3.8 kV at In 3 kA (calculated) and is monitored: reinforced is dimensioned for 6 kV (AMC3302 6.25 kV, AMC3330 7.7 kV pass on this case only). IC–04 OPEN: the G7L coil–contact rating (4000 V AC) is below reinforced – fix in rev F1 (own isolated coil supply).

SYS–IO–AUX: K_DISCH energised first at power–up; the port block itself inhibits the discharge of a closed port. Coil flyback never reaches the DO clamps in normal operation (two–switch coil drives, varistor / zener across each coil).

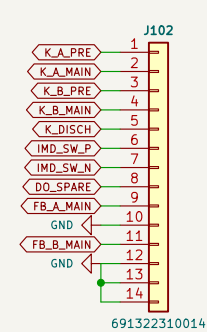
Not bench–validated. Values: sim/port_design.py → port_spec.json		
Sheet 1		
Schematics authored by Chinmoy Bhuyan		
HV–PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1); HV–PE basic + bonding		
Sheet: /		
File: PV–PORT.kicad_sch		
Title: PV–PORT 135 A port board – Cover / Hierarchy		
Size: A3	Date: 2026–10–04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 1/7

PORT CONNECTOR (CTRL-C2000) + ID WITH HOLD READBACK



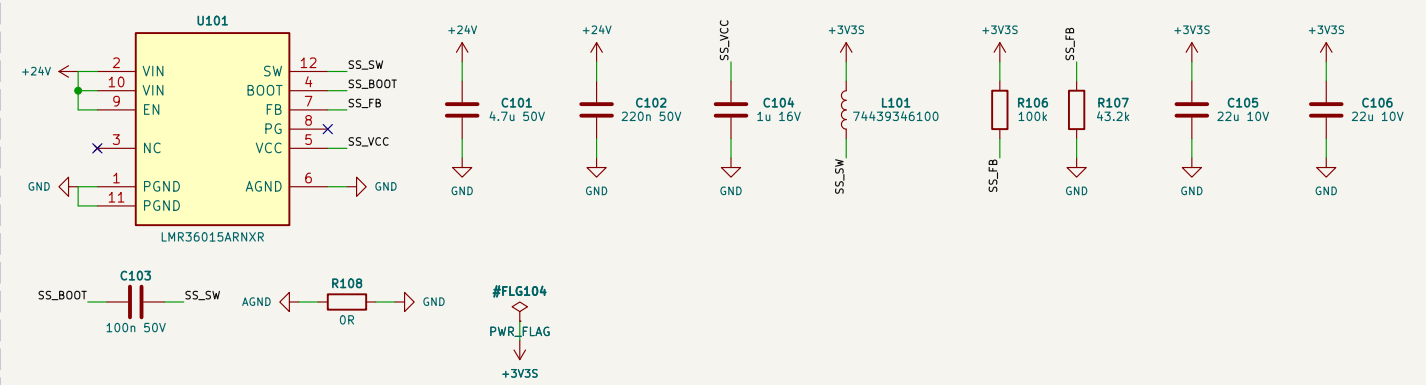
interfaces.PORT, ribbon 1:1. AMC outputs drive the cable directly (≤ 500 pF, AMC3330 p4); anti-alias RC is on CTRL-C2000. ID: 28k || readback network = 9.99 k (PV=PORT code 10.0k; CTRL-C2000 rev D values). RB outputs through 2 x 118k (port A) and 2 x 42.2k (port B): nine states, ≥ 155 counts apart (81 with the hold coils on, sim/port_design.py sec. 7.4): CTRL decodes them with its port-only table.

CONTROL CONNECTOR (SYS-IO-AUX)



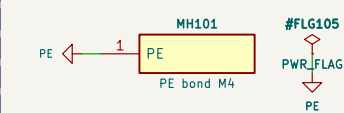
port.CTRL_PINS: DO1-DO8 (DO_SPARE = proof test), FB_A + its GND, FB_B + its GND (INT-02), 2 coil returns. Port B (battery) drives 2 HVC43 coils: K_B_MAIN up to 0.60 A -> SYS DO4 needs ILIM 10 k (1.58-2.30 A).

SENSOR SUPPLY +3V3S



LMR36015 400 kHz, 24 V $\rightarrow$ 3.315 V (100 k / 43.2 k, SNVSB49D sec 10.2), $\leq$ 0.34 A for 6 x AMC3330 + 2 x AMC3302 + hold logic. AGND joins GND only here (0 ohm).

PE BOND



Board PE to chassis at the SPD / Y-capacitor end.

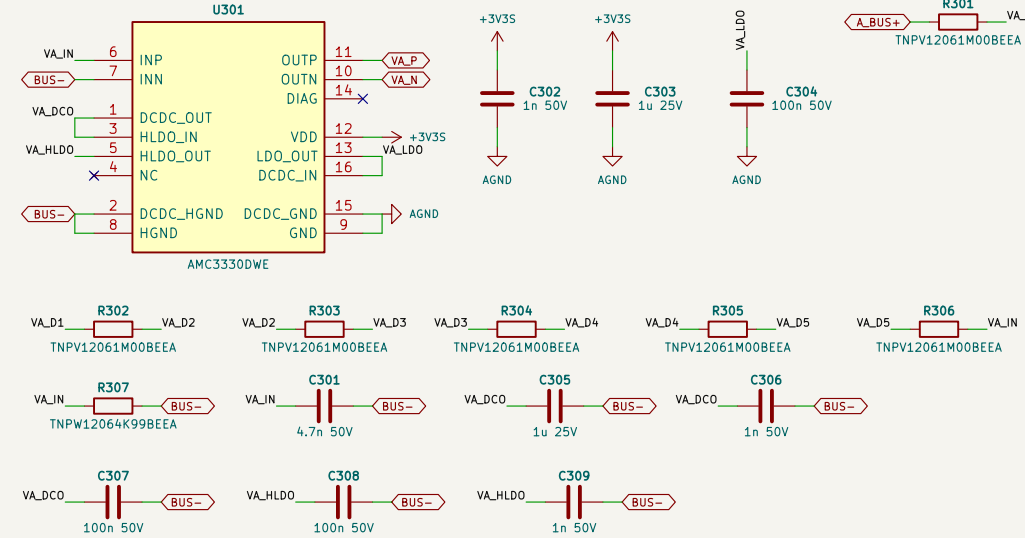
Not bench-validated. Values: `sim/port_design.py -> port_spec.json`
Sheet 2
Schematics authored by Chinmoy Bhuyan
HV-PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1); HV-PE basic + bonding

Sheet: /01 Connectors and sensor supply/
File: 01_io.kicad_sch

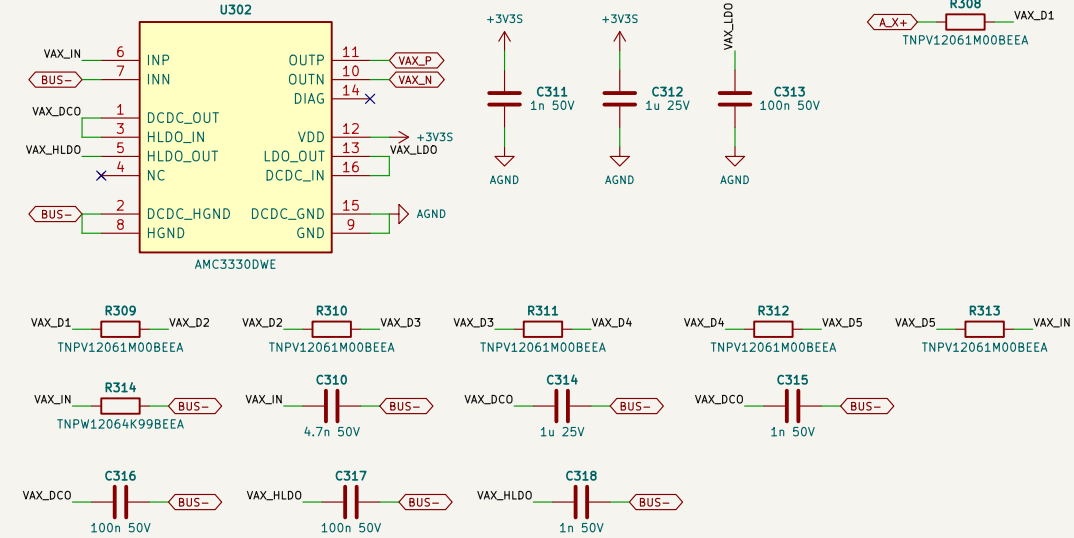
Title: PV-PORT 135 A port board – Connectors and sensor supply

Size: A3	Date: 2026-10-04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 2/7

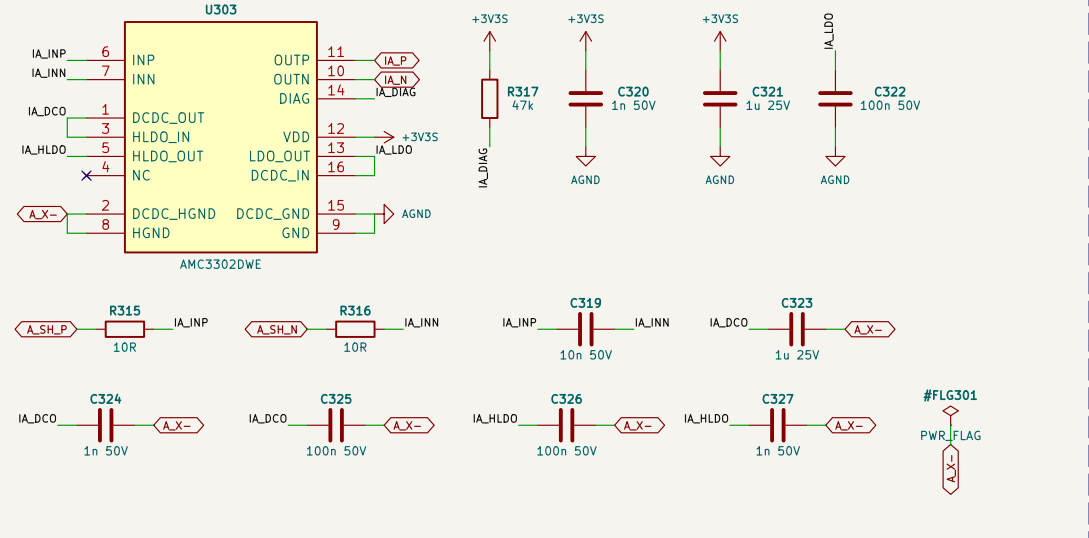
VA – BUS-SIDE VOLTAGE



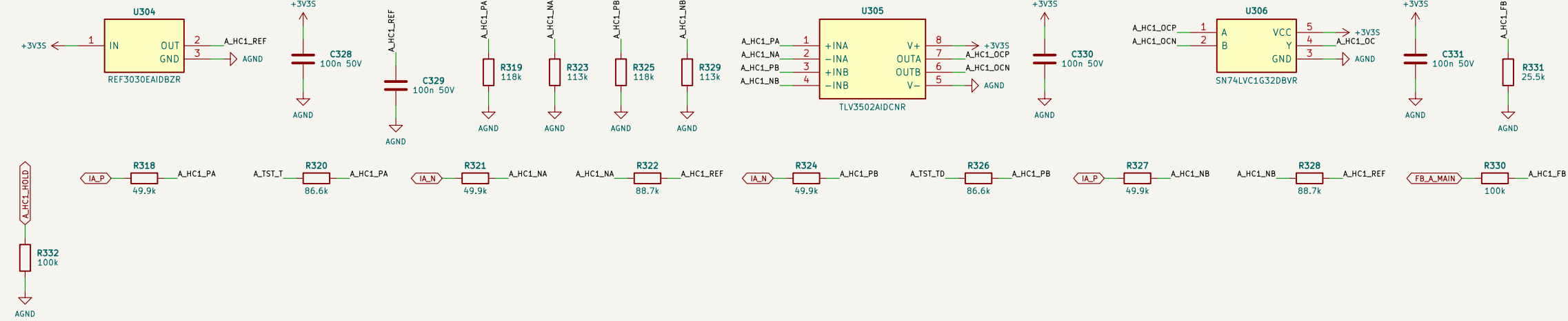
VAX – TERMINAL-SIDE VOLTAGE



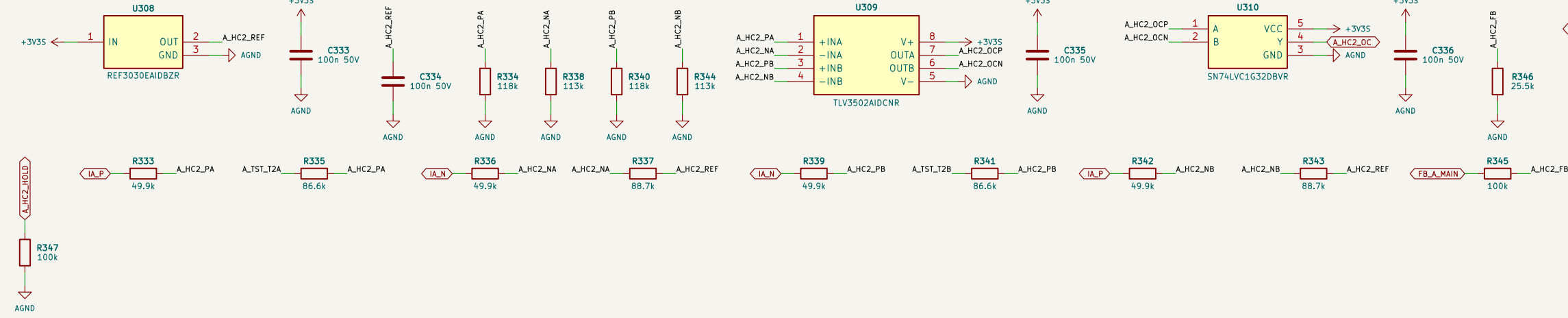
IA – PORT CURRENT



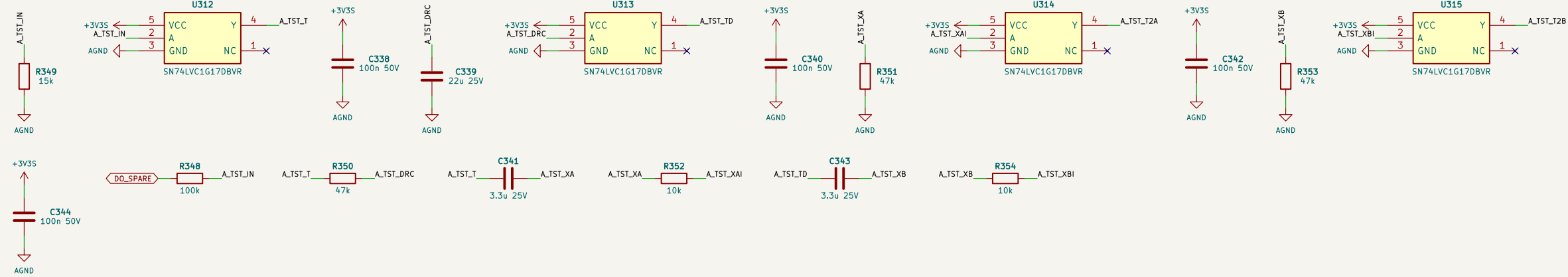
HOLD-CLOSED LOGIC, CHANNEL 1 (IRFR9214 HIGH SIDE)



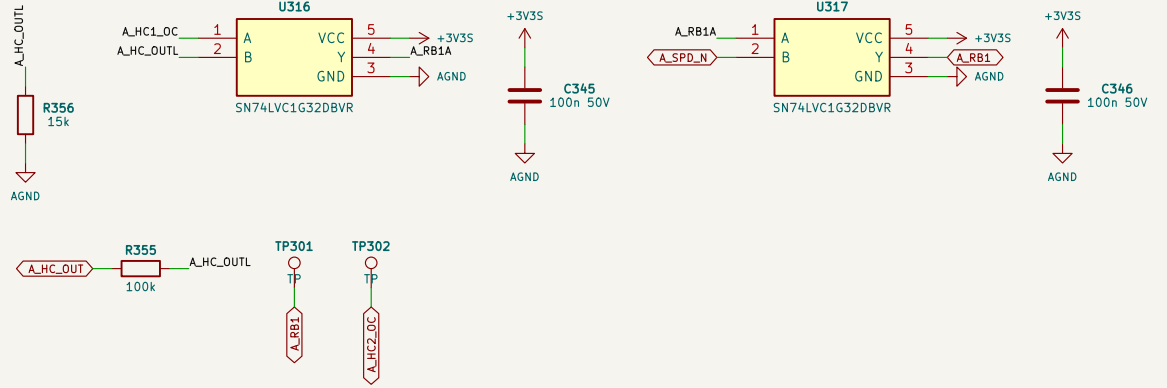
HOLD-CLOSED LOGIC, CHANNEL 2 (IRFL214 LOW SIDE)



PROOF-TEST INJECTION (DO_SPARE = TEST)



REDFBACK RB1 / RB2 (TO CTRL ON A SPARE LINE)



The schematic diagram illustrates the AMC3330DWE IC and its connections. The IC is a yellow rectangle with pins 1-16. Internal blocks include DCDC_OUT, HLDO_IN, HLDO_OUT, NC, DCDC_HGND, DCDC_GND, and GND. External components include resistors R502-R506, capacitors C502-C509, and a bus connection. Power supply pins are connected to +3V3 and AGND. Signal pins are connected to B.BUS+ and VB.D1. The diagram also shows the connection of the IC to a TNPV12061M00BEEA component.

IC Pin Connections:

- Pin 1: VB_IN (connected to B.BUS+)
- Pin 2: DCDC_HGND (connected to GND)
- Pin 3: VB_DCO (connected to GND)
- Pin 4: HLDO_IN (connected to GND)
- Pin 5: HLDO_OUT (connected to GND)
- Pin 6: INP (connected to GND)
- Pin 7: OUTP (connected to GND)
- Pin 8: INN (connected to GND)
- Pin 9: DIAG (connected to GND)
- Pin 10: VB_P (connected to GND)
- Pin 11: VB_N (connected to GND)
- Pin 12: VDD (connected to +3V3)
- Pin 13: LDO_OUT (connected to +3V3)
- Pin 14: LDO_IN (connected to +3V3)
- Pin 15: DCDC_IN (connected to GND)
- Pin 16: GND (connected to GND)

External Components:

- Resistors: R502, R503, R504, R505, R506
- Capacitors: C502, C503, C504, C505, C506, C507, C508, C509
- Bus: B.BUS+

Power Supply Connections:

- +3V3: Connected to pins 12, 13, 14, 15, 16
- AGND: Connected to pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

Signal Connections:

- B.BUS+: Connected to pin 1
- VB.D1: Connected to pin 1

IC Connections:

- IC1: TNPV12061M00BEEA
- IC2: TNPV12061M00BEEA
- IC3: TNPV12061M00BEEA
- IC4: TNPV12061M00BEEA
- IC5: TNPV12061M00BEEA

Component Values:

- R502, R503, R504, R505, R506: 1.00 M
- C502, C503, C504, C505, C506, C507, C508, C509: 1.00 nF

Notes:

- 6 x TNPV1206 1.00 M (167 V each at 1000 V) + 4.99 k: 1 V = 1203 V, clip 1504 V, 4.7 nF: lag
- ac = 27 as incl. AMC3330. Out: VB_P/N (+/-2 V on 1.44 V, PORT contract), HGND = BUS-.

Figure 10 is a schematic diagram of the TNPV12061M00BEEA board. The central component is the U502 chip, labeled AMC3330WE. The diagram shows the internal connections of the board, including the power supply, ground, and various passive components. The board is labeled 'TNPV12061M00BEEA'.

The schematic shows the following components and connections:

- U502 (AMC3330WE):** The central chip with pins 1-16. Pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 are labeled. Pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 are labeled. Pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 are labeled.
- Power Supply:** +3V3 and AGND connections. The +3V3 supply is connected to pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16. The AGND connection is connected to pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16.
- Passive Components:** Resistors (R509, R510, R511, R512, R513, R514) and capacitors (C510, C511, C512, C513, C514, C515, C516, C517, C518) are shown. The capacitors are connected to the power supply and ground.
- Board Labels:** The board is labeled 'TNPV12061M00BEEA'.

[illegible][illegible]

The schematic diagram illustrates the internal circuitry of the B-TST module, featuring four channels (U512, U513, U514, U515) and their associated components. Each channel is represented by a yellow box with pins labeled VCC, A, GND, Y, and NC. The components are connected as follows:

- Channel U512:** VCC is connected to +3V3S through a 5k resistor (R549). A is connected to B_TST_IN through a 2k resistor (R548). GND is connected to AGND. Y is connected to B_TST_T through a 4k resistor (R550). NC is connected to AGND through a 1k resistor (R552).
- Channel U513:** VCC is connected to +3V3S through a 5k resistor (R549). A is connected to B_TST_IN through a 2k resistor (R548). GND is connected to AGND. Y is connected to B_TST_T through a 4k resistor (R550). NC is connected to AGND through a 1k resistor (R552).
- Channel U514:** VCC is connected to +3V3S through a 5k resistor (R549). A is connected to B_TST_IN through a 2k resistor (R548). GND is connected to AGND. Y is connected to B_TST_T through a 4k resistor (R550). NC is connected to AGND through a 1k resistor (R552).
- Channel U515:** VCC is connected to +3V3S through a 5k resistor (R549). A is connected to B_TST_IN through a 2k resistor (R548). GND is connected to AGND. Y is connected to B_TST_T through a 4k resistor (R550). NC is connected to AGND through a 1k resistor (R552).

Additional components include capacitors C538, C539, C540, C541, C542, C543, C544, and C545, which are connected to the VCC and GND pins of the channels. The diagram also shows the connection of the B-TST module to the B-TST_T and B-TST_XA pins of the B-TST module.

100 k / 15 k -> Schmitt buffer T (ch1 side A), T -> 47 k / 22 uF -> T_d (ch1 side B). One-shots (5.5 uS / 47 k) of T -> T2A (ch2 side A) and of T_d -> T2B (ch2 side B). No single fault trips both channels for longer than one one-shot. Firmware runs it at every start, contactor open and closed (report sec. 7.4; windows, what it finds and what it does not).

RB1 = OC1 OR hold high-side output present (100 k / 15 k); at rest RB1 = 1 means a stuck channel 1.
 RB2 = OC2. PVR-PWR sums them onto the ID line (board.io); firmware reads RB1 + RB2 against the measured current in operation and through the proof-test windows at start.

Not bench-validated. Values: sim/port_design.py -> port_spec.json

Sheet 6

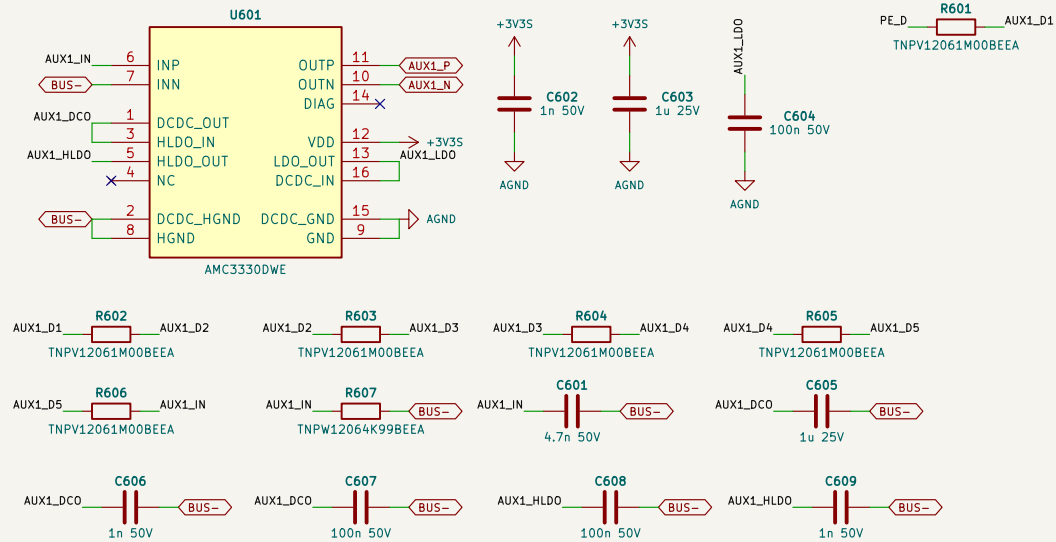
Schematics authored by Chinmoy Bhuyan

HV-PV reinforced (1000 VDC, OVC II, PD2, coated to PD1): HV-PV basic + bonding

Sheet: /05 Port B sensing and hold logic/
File: 05_portB_sense.kicad_sch

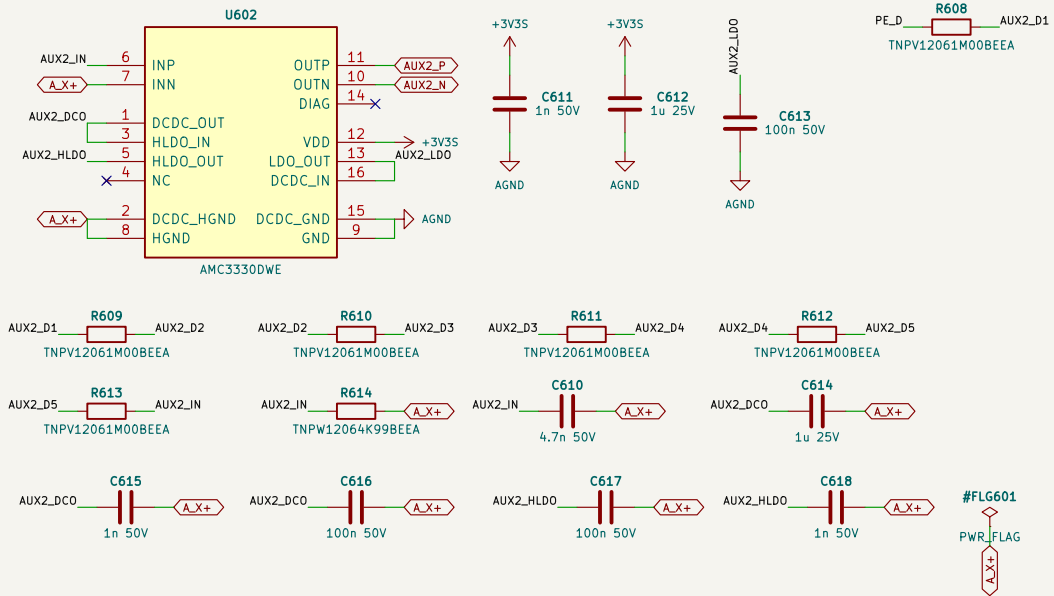
Title: PV-PORT 135 A port board – Port B sensing and hold logic		
Size: A2	Date: 2026-10-04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 6/7

AUX1 – PE RELATIVE TO BUS–



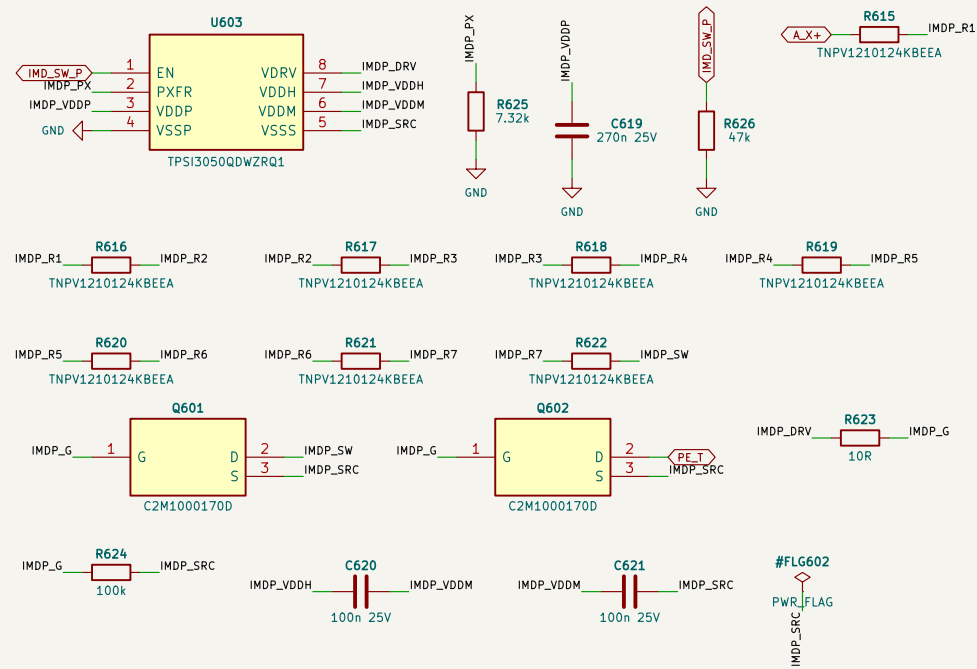
6 x 1.00 M from PE + 4.99 k: +/-1203 V linear. The divider is a fixed, known 6.005 M path to PE (included in the IMD equations, sim/port_design.py sec. 8). First element = HV-PE crossing.

AUX2 – PE RELATIVE TO A_X+



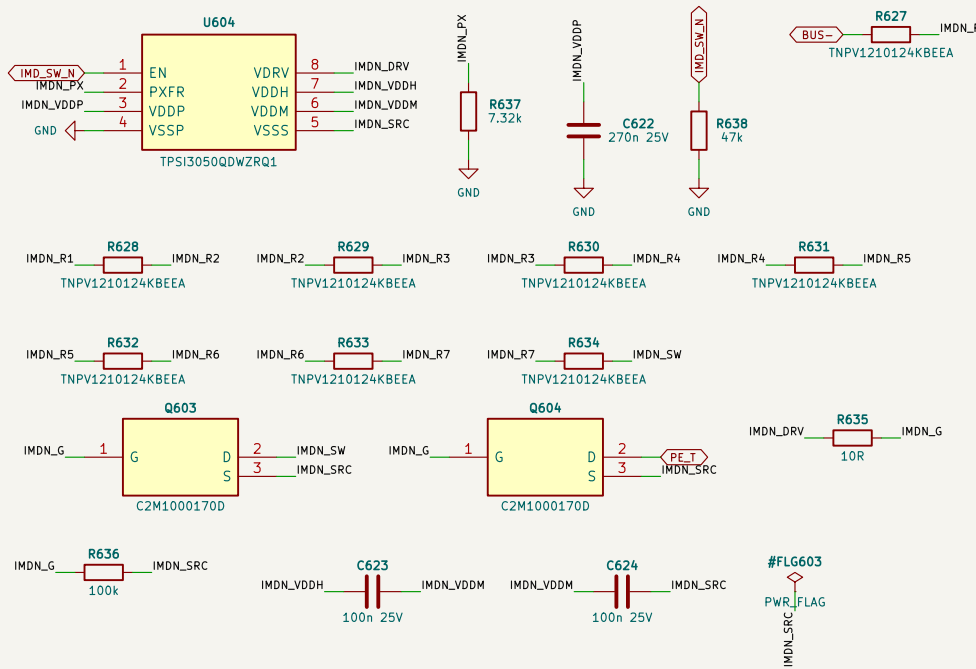
6 x 1.00 M from PE + 4.99 k: +/-1203 V linear. The divider is a fixed, known 6.005 M path to PE (included in the IMD equations, sim/port_design.py sec. 8). First element = HV-PE crossing.

TEST STRING A_X+ -> PE (IMD_SW_P)



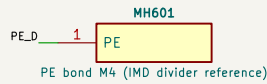
8 x TNPV1210 124 k = 992 k (125 V, 0.13 W each at 1000 V): 1 mA test current. Energised = connected. Common-source C2M1000170D pair (PA-09): blocks either polarity, so a pole below PE cannot leak through a body diode. TPSI3050 two-wire from the 24 V DO, reinforced. IDSS <= 100 uA each - EOL baseline cal.

TEST STRING BUS- -> PE (IMD_SW_N)



8 x TNPV1210 124 k = 992 k (125 V, 0.13 W each at 1000 V): 1 mA test current. Energised = connected. Common-source C2M1000170D pair (PA-09): blocks either polarity, so a pole below PE cannot leak through a body diode. TPSI3050 two-wire from the 24 V DO, reinforced. IDSS <= 100 uA each - EOL baseline cal.

PE_D BOND + ONE ACTIVE MONITOR PER DC SYSTEM



PE_D: own M4 bond (dividers only). The module disables its own measurement by leaving IMD_SW_P/N de-energised (strings open, fail-safe): a passive module still loads each pole with 6.005 M to PE.

Not bench-validated. Values: sim/port_design.py -> port_spec.json
Sheet 7
Schematics authored by Chinmoy Bhuyan
HV-PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1); HV-PE basic + bonding

Sheet: /06 Insulation measurement/
File: 06_imd.kicad_sch

Title: PV-PORT 135 A port board – Insulation measurement

Size: A3 Date: 2026-10-04 Rev: F0
KiCad E.D.A. 10.0.6 Id: 7/7